Autoregressive Streamflow Forecasting across 1–180 Day Horizons: Predictability Decay, Cross-Basin Pooling, and External Memory

A 669-basin CAMELS study of XGBoost and LSTM model families

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Abstract

Deep-learning rainfall–runoff research on CAMELS basins evaluates almost exclusively forcing-only, single-step simulation. Operational forecasting, however, is *autoregressive*: past discharge is available as input. We benchmark per-basin and pooled (regionally trained) XGBoost and LSTM variants on 669 CAMELS basins (1990–2009) under an autoregressive, leakage-safe protocol at lead times $h \in \{1, 3, 7, 10, 15, 30, 90, 180\}$ days, anchored by persistence and day-of-year climatology baselines. We map the full predictability decay curve, test whether regional training dominates per-basin calibration (Kratzert et al., 2024) in this protocol, and quantify how feature engineering and data augmentation act as *external memory* for stateless tree ensembles. [RESULTS PLACEHOLDER – final numbers to be inserted after the experiment queue completes.]

1 Introduction

Reliable streamflow prediction underpins flood early warning, reservoir operation, and water-supply planning. The task spans three regimes with distinct information sources: *short-range* forecasting (1–7 days), where catchment storage makes discharge strongly autocorrelated; *medium-range* forecasting (10–30 days), where skill must be extracted from slow storage anomalies; and *seasonal* prediction (90–180 days), where predictability rests almost entirely on initial states — chiefly snowpack and groundwater — plus climatology. Most machine-learning hydrology research addresses only the first regime; how model skill decays across the full spectrum, and whether that decay is climate-dependent, remains unmapped.

Two modeling paradigms dominate data-driven hydrology. Gradient-boosted tree ensembles are stateless function approximators whose temporal competence must be supplied externally, through windowed inputs or engineered lag/state features. Long Short-Term Memory (LSTM) networks carry state internally and can, in principle, learn storage–discharge dynamics themselves. Whether the second paradigm's structural advantage grows, shrinks, or inverts as the forecast horizon extends — and whether cross-basin training changes this balance — is an open empirical question with direct practical consequences.

2 Background and Related Work

2.1 Large-sample hydrology and CAMELS

The CAMELS dataset [Newman et al., 2015] established the standard testbed for data-driven rainfall–runoff research, extended globally by Caravan [Kratzert et al., 2023] and critically assessed

by Clerc-Schwarzenbach et al. [2024]. Calibrated outputs of process models accompany the dataset; the lumped conceptual model GR4J [Perrin et al., 2003] remains a widely used parsimonious baseline.

2.2 Deep learning rainfall–runoff

Kratzert et al. [2018] showed a single LSTM trained across hundreds of CAMELS basins outperforms calibrated conceptual models; catchment-aware variants [Kratzert et al., 2019] generalize further, and Kratzert et al. [2024] demonstrated that regional training dominates per-basin calibration even with long records. Gauch et al. [2021] benchmarked trees against LSTMs under varying data volume; Frame et al. [2022] addressed extremes; Schoppa et al. [2020] characterized random forests in large-sample hydrology.

2.3 Operational multi-horizon forecasting

Google's global flood system extended reliable lead times to 7 days [Nearing et al., 2024]. Hunt et al. [2022] showed LSTM skill persists to ∼10 days; beyond that, studies report steep degradation [Li et al., 2023, Islam et al., 2025]. Seasonal prediction is traditionally served by physically-based procedures; observational work on storage–discharge dynamics [Kirchner, 2009] explains why initial storage carries multi-month memory in snow-dominated catchments. No existing work traces a single framework across the full 1–180 day range.

2.4 The gap

(i) The CAMELS line evaluates almost exclusively forcing-only, single-step simulation; the autoregressive protocol is rarely benchmarked at scale. (ii) Whether regional>local transfers to autoregressive forecasting, and whether stateless trees benefit from pooling differently than recurrent networks, is untested. (iii) Feature engineering and data augmentation act as external memory for stateless models; no study quantifies how their value interacts with pooling and horizon.

3 Data and Task Definition

669 CAMELS basins (671 minus two with NaN/inf records), 1990-01-01 to 2009-12-31; seven daily drivers (daylength, precipitation, shortwave radiation, max/min temperature, vapor pressure, discharge). Chronological 80/20 split per basin (train ≈ 1990–2005.9, test 2005.10–2009); normalization statistics fitted on the train segment only.

Task. Direct multi-horizon forecasting: inputs use only data up to day t; the target is discharge at day $t + h$, $h \in \{1, 3, 7, 10, 15, 30, 90, 180\}$. Input window $w(h)$: 15 ($h \leq 7$), 60 ($h \in \{10, 15\}$), 90 ($h = 30$), 270 ($h \in \{90, 180\}$). No recursion, no future forcing. Baselines: persistence (reported $h \leq 7$) and day-of-year climatology (training-period doy mean, 11-day centered smoothing, reported at all h).

4 Models

4.1 Per-basin family

XGBoost on flattened raw windows (`xgb-raw`) or engineered same-day feature vectors (`xgb-fe`: lags, antecedent precipitation indices, rolling sums, snow/energy proxies, seasonality harmonics,

Table 1: Predictability decay (median NSE; h≤15: 669 basins, h=30: 335, h=90/180: 34 stratified basins).

Track	h=1	h=3	h=7	h=10	h=15	h=30	h=90	h=180
persistence	0.494	−0.148	−0.465	−0.545	−0.603	−0.751	−1.071	−1.173
climatology	–	–	–	0.019	0.019	0.018	0.037	0.061
xgb-fe	0.583	0.190	0.088	0.067	0.056	0.032	0.012	0.019
lstm-raw-zscore	0.609	0.253	**0.151**	**0.133**	**0.113**	**0.074**	**0.032**	**0.046**
lstm-pooled-emb	**0.646**	**0.269**	0.150	0.098	0.066	0.018	0.014	−0.001

Table 2: Pooling reversal (paired per-basin; FDR-significant where marked).

h	median ΔNSE	wins	Cohen's d	FDR
1	+0.028	589/669	0.83	yes
3	+0.009	502/669	0.36	yes
7	+0.004	417/669	0.00	yes
10	−0.013	214/669	−0.30	no
15	−0.026	166/669	−0.43	no
30	−0.037	85/335	−0.31	no
90	−0.018	–	–	no
180	−0.047	–	–	no

discharge memory); LSTM (hidden 64, 2 layers, SmoothL1, Adam, StepLR) with Z-Score or Min-Max per-basin normalization; `-fe` variants append engineered state channels; `-fe-long` adds 90-day memory channels for $h \geq 30$.

4.2 Pooled family

One shared model trained on all basins' train segments (test segments held out basin-wide): `pooled-plain` (no station identifier), `pooled-emb` (+learned 16-dim station embedding broadcast to every timestep), `pooled-emb-fe`, and XGBoost with station integer code as a split feature. Note: embeddings encode learned per-basin capacity, not physical attributes (the CAMELS attributes file is outside our data package).

4.3 Physics reference

GR4J [Perrin et al., 2003] with Oudin PET, differential-evolution calibration on the train segment (h=1 simulation mode), as a research replication; CAMELS official calibrated model outputs serve as the literature reference.

5 Experimental Design

[TUNING/AUGMENTATION/METRICS/STATISTICS – carried from docs/BACKGROUND.md sections 5–6; Optuna TPE search protocol; bootstrap 95% CIs, BH-FDR, rank-biserial and Cohen's d.]

Table 3: State–capacity axis (335-basin subset).

Track	h=1	h=7	h=30
mlp-zscore	0.626	0.136	−0.014
attn-zscore	0.587	0.138	0.076
gru-zscore	0.618	0.150	0.079
lstm-raw-zscore	0.609	0.151	0.074

6 Results

Finding 1 (decay + cliff). Median-basin skill is nearly exhausted by h=30 (LSTM 0.074), yet 307 snow-dominated basins retain NSE $\approx$ 0.10 at h=180 — indistinguishable from day-of-year climatology (0.103) — while rain basins collapse to $\approx$0.02. Seasonal predictability is initial snowpack memory, and its limit is climate-type-dependent.

Finding 2 (pooling reversal). Pooled LSTM beats per-basin at h=1 (589/669, d=0.83, FDR-sig), ties at h=7, and loses from h=10 onward, deepening to −0.047 at h=180. Pooled XGBoost is worse at every horizon and collapses at seasonal horizons (−0.18 at h=180). Cross-basin training transfers universal short-range dynamics, not basin-specific long-range storage behavior.

Finding 3 (state–capacity). A stateless MLP matches LSTM at h=1 and collapses by h=30; GRU $\approx$ LSTM; attention tracks LSTM through mid-range. Recurrent state is the load-bearing structure exactly where the window stops being sufficient.

Finding 4 (external memory is a no-op). Feature engineering shows no paired gain at any horizon (all FDR n.s.), including 90-day memory channels; augmentation and 1.0σ test-noise probes move NSE <0.01 — the discharge channel dominates the input information.

Finding 5 (normalization + tuning). Z-Score beats Min-Max on 669/669 basins (median ΔNSE +0.29); Optuna tuning adds +0.03. The largest levers in this protocol are trivial choices, not architecture.

Physics reference. The official NWS HL-RMS calibrations (snow-aware, Daymet forcing), evaluated under our protocol with train-only variant selection (438 gauges), reach median NSE 0.649 at h=1 (snow 0.633 / rain 0.672) — the strong simulation anchor that our GR4J replication (no snow module, −0.30) falls short of. HL-RMS is a forcing-only simulation; the comparison with our autoregressive forecasts is indicative, not protocol-matched.

7 Discussion and Limitations

[To write after results.]

8 Conclusion

[To write after results.]

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